

ERDENETSAGAAN, Mongolia

WMO: 443170

Lat: 45.9067N

Lon: 115.3719E

Elev: 3530

StdP: 12.92

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.4	-17.5	-27.4	1.4	-21.0	-23.8	1.7	-17.0	31.8	1.7	27.6	1.0	8.3	320	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	89.5	66.2	85.4	64.0	81.8	62.5	68.9	84.5	66.4	80.7	64.4	78.2	10.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	101.1	76.6	61.3	92.5	73.3	59.2	85.8	71.5	35.5	84.5	33.2	80.9	31.5	78.2	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.1	26.4	22.1	-26.6	94.5	3.6	4.7	-29.2	97.8	-31.4	100.6	-33.4	103.2	-36.1	106.6
			-26.4	71.7	3.2	3.3	-28.6	74.1	-30.5	76.1	-32.3	77.9	-34.6	80.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.6	-0.8	5.2	20.9	38.7	52.4	63.4	69.0	65.7	54.0	37.5	17.0	2.4	(6)
(7)		DBStd	26.49	9.42	10.76	11.91	10.19	9.31	7.57	5.97	6.81	8.63	10.17	11.55	9.40	(7)
(8)		HDD50	7102	1574	1255	901	361	82	2	0	1	56	405	990	1475	(8)
(9)		HDD65	11056	2039	1675	1366	788	402	117	24	72	340	853	1440	1940	(9)
(10)		CDD50	1857	0	0	0	22	158	405	590	488	177	17	0	0	(10)
(11)		CDD65	337	0	0	0	0	13	70	149	94	11	0	0	0	(11)
(12)		CDH74	3593	0	0	0	7	236	790	1413	1000	145	2	0	0	(12)
(13)		CDH80	1168	0	0	0	0	60	269	493	323	23	0	0	0	(13)
(14)	Wind	WSAvg	9.3	9.5	9.6	10.3	11.2	11.8	8.3	7.1	7.1	8.8	9.1	9.6	9.8	(14)
(15)	Precipitation	PrecAvg	10.0	0.1	0.1	0.2	0.3	0.7	1.5	2.7	2.5	1.2	0.3	0.2	0.1	(15)
(16)		PrecMax	16.8	0.3	0.3	0.7	1.7	2.0	3.5	7.5	5.6	3.2	1.1	0.4	0.3	(16)
(17)		PrecMin	4.5	0.0	0.0	0.0	0.0	0.0	0.2	0.6	0.4	0.1	0.0	0.0	0.0	(17)
(18)		PrecStd	2.2	0.1	0.1	0.2	0.4	0.6	0.9	1.5	1.4	0.8	0.3	0.1	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	39.3	58.2	74.1	86.8	93.0	95.4	94.3	83.5	70.7	52.3	32.8	(19)
MCWB			22.7	31.2	42.2	51.7	58.9	66.2	70.1	68.9	60.7	51.7	40.1	27.8	(20)	
2%		DB	21.5	32.5	51.2	66.9	80.3	87.0	89.7	87.8	78.3	64.7	44.7	24.7	(21)	
		MCWB	17.7	26.6	38.2	48.0	55.9	62.9	67.7	65.8	57.9	49.4	35.4	21.0	(22)	
5%		DB	16.9	27.0	44.7	61.5	75.4	82.4	85.4	83.0	74.0	58.9	38.7	20.1	(23)	
		MCWB	14.1	22.7	34.3	45.3	54.2	61.8	65.2	63.4	55.6	45.5	31.5	17.2	(24)	
10%		DB	13.0	21.8	38.6	56.5	70.3	78.0	81.6	79.5	69.6	54.2	33.7	15.9	(25)	
		MCWB	10.9	18.4	30.5	42.2	51.6	59.8	64.0	61.6	53.6	42.6	28.4	13.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	31.9	43.1	53.1	60.8	69.2	73.7	71.2	63.3	54.3	40.2	27.4	(27)
MCD B			27.7	37.7	57.1	70.5	83.0	88.7	91.4	87.9	79.1	67.7	51.5	33.1	(28)	
2%		WB	17.9	26.7	38.6	49.1	57.6	65.7	69.8	68.2	59.5	49.5	35.8	21.0	(29)	
		MCD B	21.4	32.3	50.3	65.2	77.6	82.5	84.8	84.7	74.5	63.7	44.6	24.5	(30)	
5%		WB	14.2	22.7	34.7	45.7	55.2	63.3	67.3	65.3	57.1	45.9	31.7	17.2	(31)	
		MCD B	16.8	27.1	44.0	60.4	72.7	78.5	80.6	79.6	70.8	57.7	38.5	20.1	(32)	
10%		WB	10.9	18.4	30.7	42.5	52.6	61.2	65.2	63.0	54.7	42.7	28.3	13.6	(33)	
		MCD B	13.0	21.6	38.1	56.4	68.6	74.8	78.3	76.0	67.5	53.5	33.7	16.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.1	18.5	20.2	22.3	23.2	20.5	19.6	21.4	22.1	20.9	17.9	15.2	(35)
MCD BR			21.3	23.2	25.7	28.8	29.8	25.9	24.7	25.5	26.7	26.5	22.8	19.6	(36)	
MCWBR			18.7	18.8	16.8	17.2	15.8	12.5	11.5	12.2	14.3	16.9	16.4	16.9	(37)	
5% WB		MCD BR	21.2	22.7	25.1	27.2	27.7	22.9	21.9	22.3	24.2	25.6	22.6	19.3	(38)	
		MCWBR	18.7	18.5	17.1	17.2	15.5	12.2	11.7	12.0	14.0	17.1	16.6	16.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.254	0.278	0.324	0.379	0.447	0.462	0.416	0.387	0.344	0.313	0.276	0.259	(40)	
(41)		taud	2.328	2.285	2.216	2.114	1.960	2.038	2.246	2.293	2.352	2.359	2.341	2.298	(41)	
(42)		Ebn at Noon	273	286	285	277	261	257	269	272	276	269	260	256	(42)	
(43)		Edh at Noon	23	30	38	47	56	52	42	39	33	28	23	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	678	1043	1491	1840	2059	2013	1954	1823	1496	1073	703	545	(44)	
(45)		RadStd	44	67	58	92	127	149	89	94	88	64	41	34	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (4 neighbors)	N/A	N/A	-3.13	N/A	-1.04	-1.55	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page